DO NOW — RECALL

Lesson 156 · Client-Side versus Server-Side Processing

Name: _____ Class: _____ Date: _____

1. Why does PageRank divide a page's score among outgoing links?

2. What data structure follows LIFO?

3. What is a database foreign key?

4. What is the role of a router?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Each outgoing link gets a share of source importance.

2. Model answer

Stack.

3. Model answer

A reference to a key in another table.

4. Model answer

Forward packets between IP networks.

By the end, you can…

• JavaScript can update local interface promptly.

• Server authenticates, authorises and validates record changes.

• Balance speed, privacy, load and security boundaries.

Vocabulary — know these terms

client-side processing — Computing performed in browser/device.

server-side processing — Computing performed by backend.

authentication — Checking claimed identity.

authorisation — Checking what a user may do.

NEW LEARNING

Read, annotate and explain

Browser tasks

JavaScript can update local interface promptly.

Backend tasks

Server authenticates, authorises and validates record changes.

Compare deployment

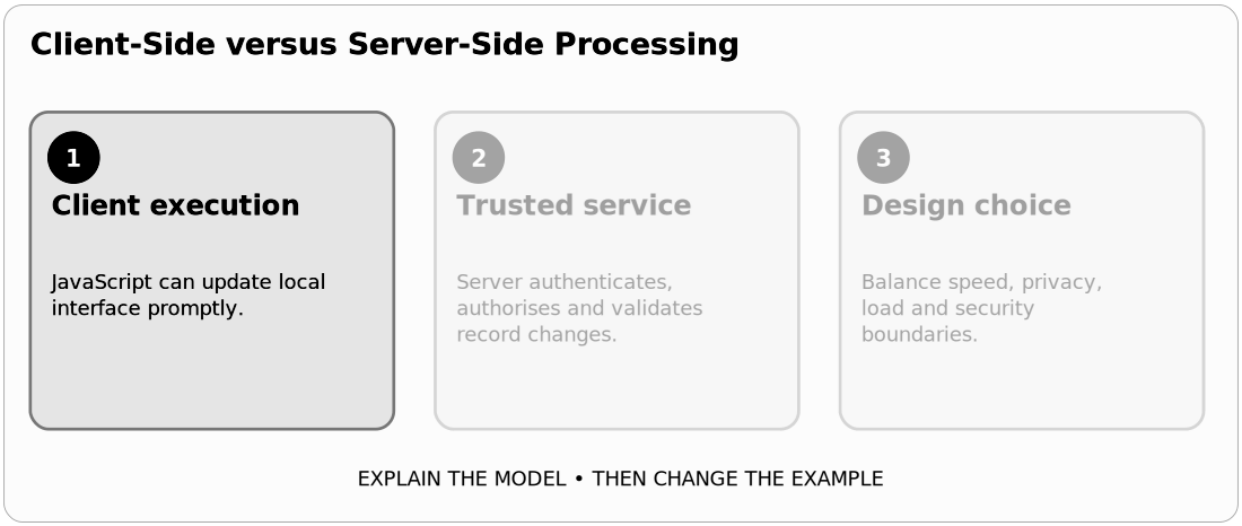
Balance speed, privacy, load and security boundaries.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

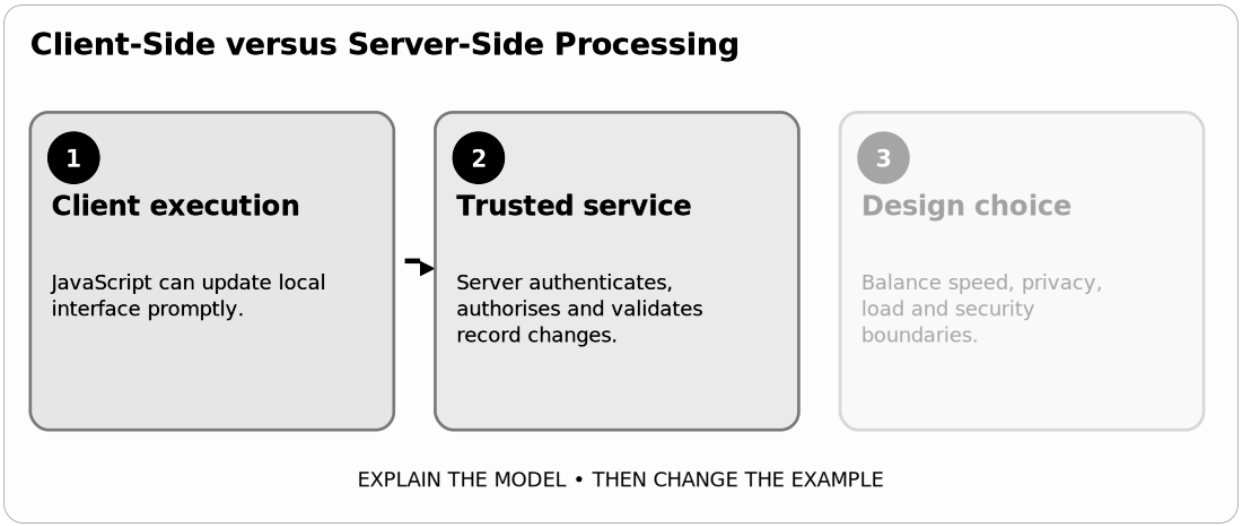
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Client-Side versus Server-Side Processing: Client execution, Trusted service, Design choice

New Learning

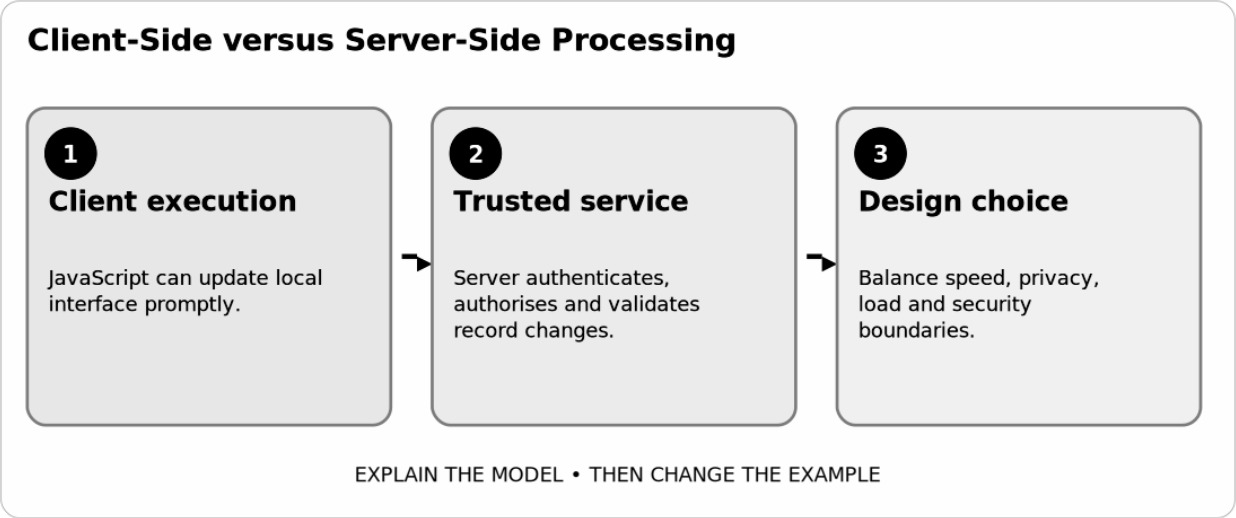


Client-Side versus Server-Side Processing: Client execution, Trusted service, Design choice

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Client-Side versus Server-Side Processing: Client execution, Trusted service, Design choice

I DO — WORKED EXAMPLE

Follow the worked method

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system protected?

Browser value is untrusted

Identify the input or diagram and predict the first result.

Server checks permissions

Follow the second stage and explain its importance.

Reject forbidden edit

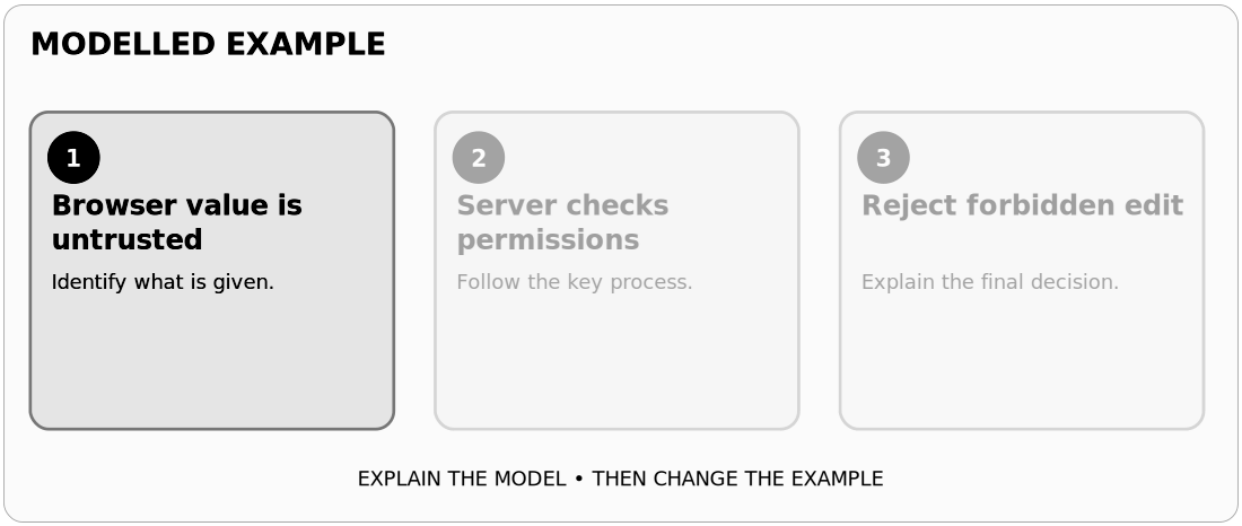
The server must authenticate the user and enforce edit permissions, ignoring unauthorised mark values. It validates requests and writes to the authoritative database only after checking access.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system p

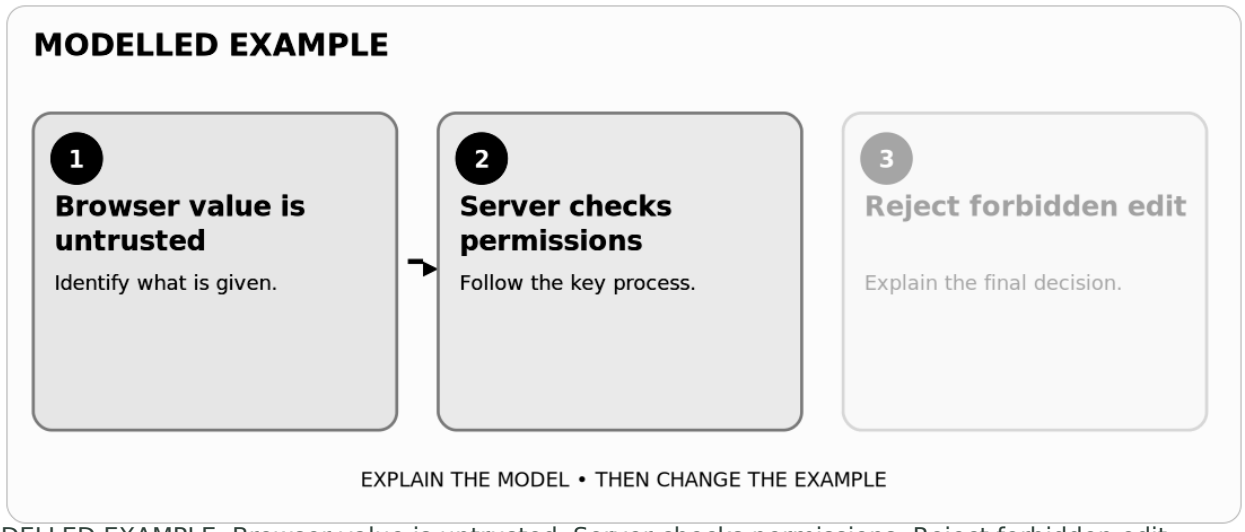
Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: Browser value is untrusted, Server checks permissions, Reject forbidden edit

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system p

Identify the input or diagram and predict the first result.



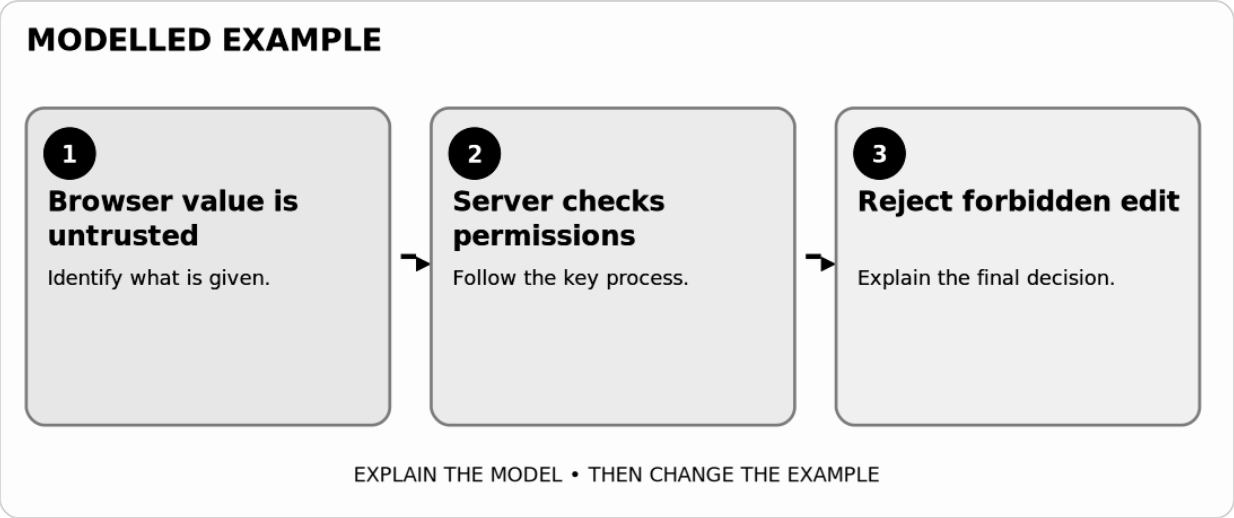
MODELLED EXAMPLE: Browser value is untrusted, Server checks permissions, Reject forbidden edit

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A pupil edits a teacher-assigned mark in browser developer tools before submitting a form. How is the system p

Identify the input or diagram and predict the first result.



MODELLED EXAMPLE: Browser value is untrusted, Server checks permissions, Reject forbidden edit

WE DO — GUIDED PRACTICE

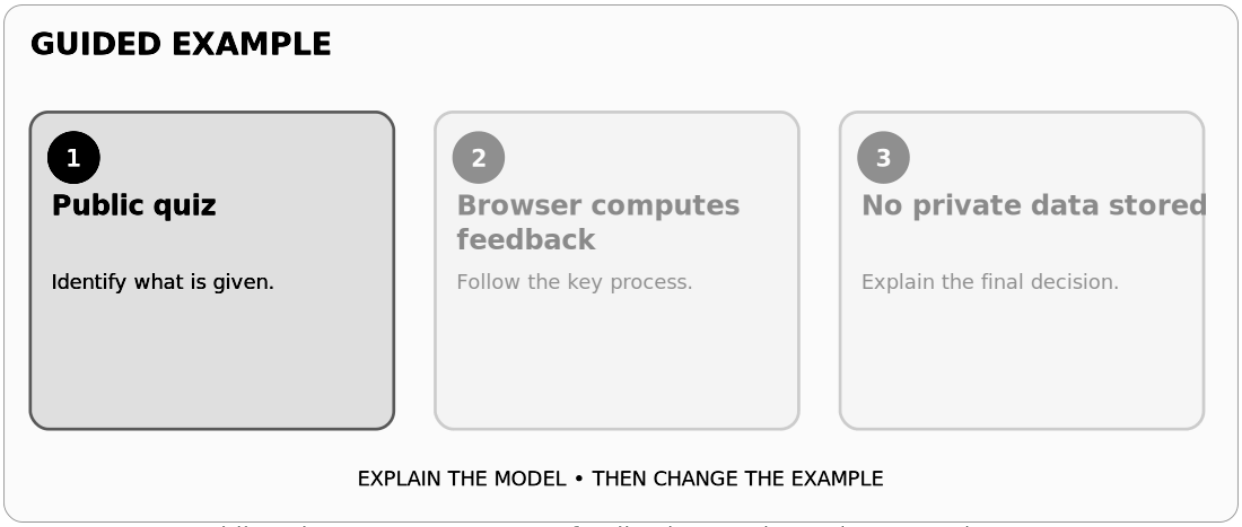
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A public revision quiz shows hints instantly but stores no marks. Which processing placement fits?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Public quiz, Browser computes feedback, No private data stored

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Public quiz Identify the input or diagram and predict the first result.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give two benefits and one limitation of client-side processing. [6]

Question 2

Explain why permission to change assessment results must be checked server-side. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A hub offers instant quiz hints, private reports and offline revision. Assign client/server roles and justify. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Browser validation is sufficient to authorise database changes.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

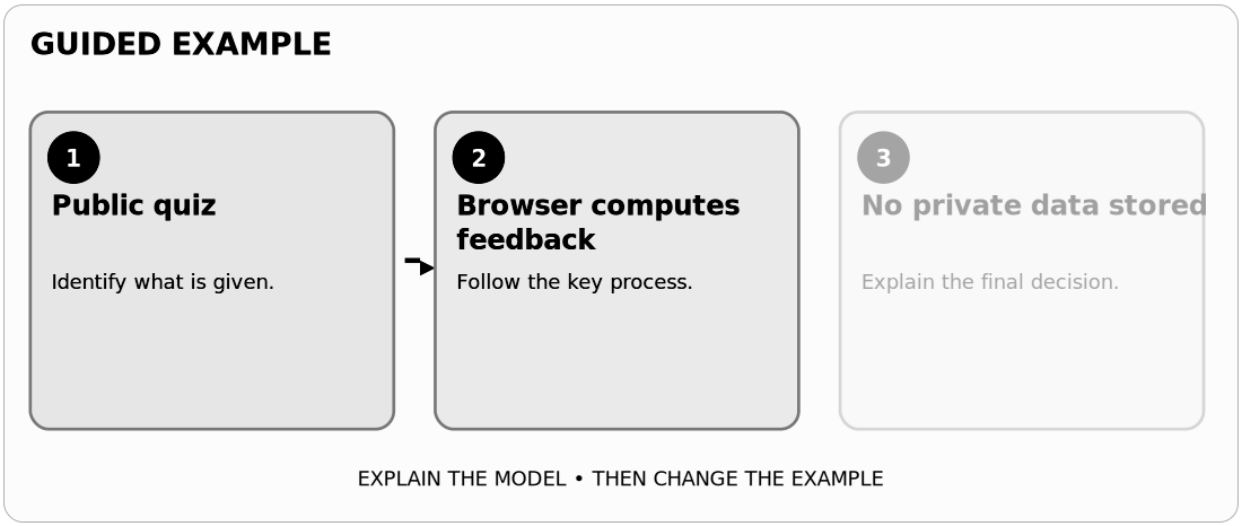
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A public revision quiz shows hints instantly but stores no marks. Which processing placement fits?

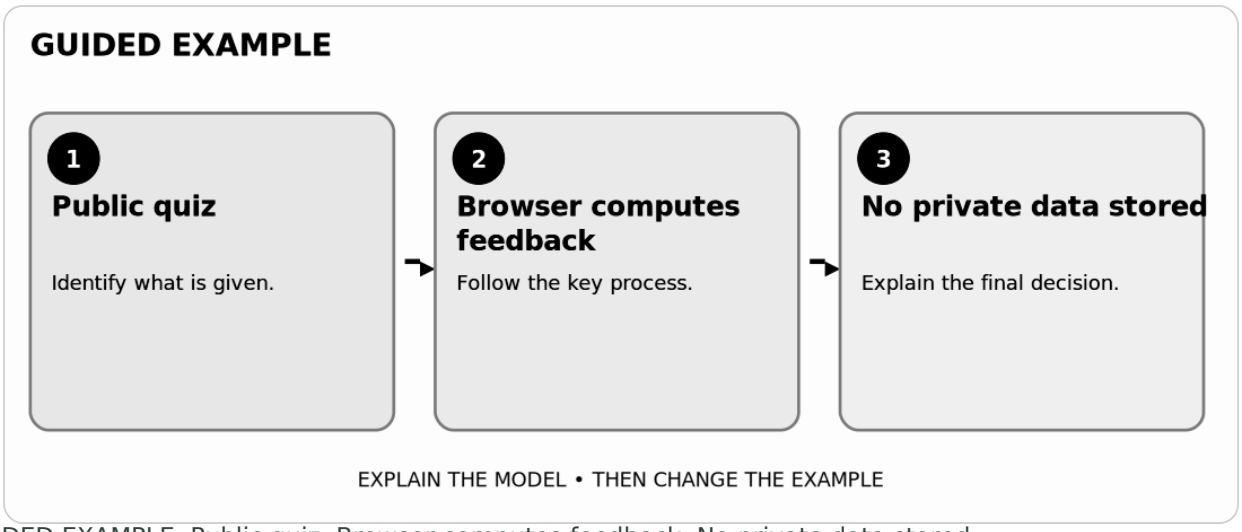
Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Public quiz, Browser computes feedback, No private data stored

A public revision quiz shows hints instantly but stores no marks. Which processing placement fits?

Identify the input or diagram and predict the first result.



GUIDED EXAMPLE: Public quiz, Browser computes feedback, No private data stored

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Browser computes feedback

Follow the second stage and explain its importance.

No private data stored

Client-side JavaScript can show hints immediately with fewer server requests; if restricted answers or persistent records matter, keep them server-controlled.

IMPROVE: Explain one correction to your own reasoning.